**Q-A device in your simulated network is not receiving an IP address from the DHCP server and shows an APIPA address (169.x.x.x). List the possible causes for this and document the step-by-step process you would use to resolve it.

Hint: Check for network connectivity, DHCP scope exhaustion, and server availability.**

To identify the causes of a client receiving an Automatic Private IP Address (APIPA) instead of an IP address from the DHCP server and to resolve the issue.

Scenario

A client PC is configured to obtain an IP address automatically using DHCP. However, instead of receiving an IP address from the DHCP server, it assigns itself an **APIPA address**.

Example:

IP Address.............169.254.10.25
Subnet Mask............255.255.0.0
Default Gateway........Not Assigned

This indicates that the client was unable to contact a DHCP server.

Possible Causes

1. The network cable is disconnected or the switch port is down.
2. The router or DHCP server is powered off or unavailable.
3. DHCP service is disabled.
4. Incorrect DHCP pool configuration.
5. DHCP address pool (scope) is exhausted (no IP addresses available).
6. Incorrect VLAN or switch port configuration preventing communication with the DHCP server.
7. Router interface is administratively down (shutdown).
8. If the DHCP server is on a different network, the router is missing the ip helper-address configuration.

Troubleshooting Steps

Step 1: Check Physical Connectivity

- Ensure the Ethernet cable is connected.
- Verify that the switch and router interfaces are active (green in Packet Tracer).

Step 2: Verify Client Configuration

On the PC:

- Desktop → IP Configuration
- Confirm **DHCP** is selected.

Run:

ipconfig

Example output:

IP Address.............169.254.10.25
Subnet Mask............255.255.0.0
Default Gateway........Not Assigned

This confirms the client failed to obtain a DHCP lease.

Step 3: Verify Router Interface

On the router:

show ip interface brief

Example:

Interface IP Address Status
FastEthernet0/0 192.168.1.1 up

If the interface is **administratively down**, enable it:

configure terminal
interface fa0/0
no shutdown
end

Step 4: Verify DHCP Configuration

Check the DHCP configuration:

```
show running-config
```

Confirm that:

- The DHCP pool exists.
- The correct network is configured.
- A valid default gateway is configured.

Example:

```
ip dhcp pool LAN
 network 192.168.1.0 255.255.255.0
 default-router 192.168.1.1
 dns-server 8.8.8.8
```

Step 5: Check for DHCP Scope Exhaustion

View DHCP bindings:

```
show ip dhcp binding
```

If all available addresses are already leased, expand the DHCP pool or remove unused bindings.

Step 6: Verify DHCP Server Availability

Ensure the DHCP server or router is powered on and connected to the network.

If using a separate DHCP server, verify it has:

- Correct IP address
- Correct subnet mask
- DHCP service enabled

Step 7: Renew the Client IP Address

On the PC:

```
ipconfig /release
ipconfig /renew
```

Or in Packet Tracer:

- Change IP Configuration from **DHCP** to **Static**, then back to **DHCP**.

result:

IP Address.............192.168.1.2
Subnet Mask............255.255.255.0
Default Gateway........192.168.1.1

Solution

The issue was resolved by ensuring:

- The network connection was active.
- The router interface was enabled.
- The DHCP service and pool were correctly configured.
- The DHCP scope had available IP addresses.
- The client renewed its DHCP lease.

After these steps, the client successfully received a valid IP address from the DHCP server.

Result

The client initially received an APIPA address because it could not contact the DHCP server. After checking network connectivity, verifying the DHCP configuration, ensuring available IP addresses in the DHCP pool, and renewing the DHCP lease, the client successfully obtained a valid IP address.